
The Song, Not the Pin: Provenance-Conditioned Route Transfer across Private Frames

Abstract

Collective-memory systems allow an agent to act on evidence that was acquired by another agent, yet conventional task and stage metrics do not identify when a commitment is supported primarily by foreign memory. We introduce a provenance-conditioned extension of Q/R/M/C. For each materialized target, the method records the foreign share φ of the evidence mass used by the memory merge and freezes this value at commitment time. A semantic-warp event is then defined as a materialization whose supporting evidence exceeds a specified foreignness threshold. This annotation permits three analyses: direct completion stratified by provenance, deterministic counterfactual replay with foreign evidence removed, and a closed-form admissibility horizon derived from the trust-staleness gate. On a 2,400-episode scarcity benchmark, foreign-evidence locks complete substantially less often than own-evidence locks (0.004–0.020 versus 0.22–0.41), but this direct comparison understates their value. Lock-chain analysis shows that foreign evidence frequently transports an agent into self-confirmation range, accounting for up to 51% of subsequent own-evidence completions. The trust-staleness law predicts the latest admissible intention time exactly on the registered 6/6 hold-out. The same analysis identifies target contention, rather than incorrect evidence, as the principal completion cost of fast sharing. We therefore introduce a decentralized reservation and rollback protocol that increases success at every tested fast cadence and exceeds the strongest fixed-cadence baseline (0.614–0.620 versus 0.583). A controlled LLM experiment further demonstrates cross-session use of foreign memory. Extending provenance from target records to route edges then reveals a structural boundary between place-only and route-level transfer: place-only guidance fails at the first obstructing structure and can perform worse than unguided exploration, whereas transferred routes remain executable. An edge-specific rupture law predicts all 110 tested validity regimes and all 21 intermediate failure indices exactly, and a traveler on a foreign route preserves the witness’s measured risk profile up to the predicted rupture. Semantic landmark constellations recover correspondence across translations and quarter-turn rotations in 80/80 tests and fail closed under ambiguity. Landmarks and edges provide complementary identity and transport functions; their composition reaches an unseen target at 6.4% of the full-snapshot bit cost before protocol metadata. These results indicate that the value of shared memory depends on the provenance, temporal validity, and coordination authority attached to a record, not only on its semantic content. The scope is limited to controlled grid and continuous simulations and a small text-world demonstration.

We claim. Foreign navigation memory becomes executable only when referential identity, relational transport, evidence provenance, temporal validity, and coordination authority are preserved together. A target without a route is operationally incomplete; a route without a valid cross-frame correspondence is unsafe; and foreign evidence without a validity and coordination contract should not directly determine action. The correct unit of transferable navigation memory is therefore not a coordinate target but a provenance-conditioned relational route whose landmarks establish identity, whose edges provide transport, and whose validity

contract determines how far and under what conditions another agent may execute it.

1 Introduction

A shared memory architecture is usually described in terms of the information available to each agent. This description is incomplete. Two agents may retrieve the same target record, but the epistemic basis of their commitments can be different: one may rely on direct observation, whereas the other may rely entirely on a peer’s report. End-to-end success, retrieval accuracy, and the original Q/R/M/C rates do not distinguish these cases because they do not condition materialization and completion on evidence provenance.

This paper introduces a provenance-conditioned measurement layer for collective semantic memory. Every contribution to a merged record retains its source, support, confidence, trust, and age. For a selected candidate, the method computes the foreign share φ using the same weights that produced the merge. The value is frozen when the agent commits to the target so that later self-confirmation does not retroactively change the provenance of the decision. A semantic-warp event is therefore not an additional stage, but a stratification of materialization and completion according to the source of the supporting evidence.

The analysis separates three questions. First, how often do foreign-evidence commitments complete directly? Second, what is the counterfactual effect of foreign memory when its contribution is removed under an otherwise identical replay? Third, for how long may foreign evidence remain action-authoritative under explicit trust and staleness rules? The symbolic memory representation makes all three questions operational: the merge is source-decomposable, the simulator supports paired deterministic replays, and the exponential staleness gate yields a closed-form horizon.

The results show that direct completion is only one component of foreign-memory value. Under target scarcity, foreign locks rarely complete because several agents can commit to the same resource. Nevertheless, the same lock often guides the recipient to a region in which it can obtain its own confirming evidence. We therefore treat foreign memory as a transport mechanism as well as a destination proposal. This distinction explains why an architecture can receive a positive counterfactual benefit from foreign evidence even when $P(C^* | W^*)$ is small.

The observed failure is primarily coordinative. Fast exchange increases the number of commitments based on the same peer evidence, which improves materialization but increases duplicate target claims. To address this mechanism, we introduce a minimal decentralized protocol consisting of optimistic commitment, reservation messages, and rollback with backoff. The protocol does not alter the semantic content of the memory; it regulates when a shared record is permitted to determine action.

The contributions of this paper are as follows:

1. We define a provenance-conditioned annotation of materialized targets and completion events, together with a deterministic counterfactual measure of the contribution of foreign evidence.
2. We derive a trust–staleness admissibility horizon and validate its discrete predictions on a registered hold-out.
3. We identify downstream transport through mixed-provenance lock chains as a major component of foreign-memory value.
4. We introduce and evaluate a decentralized reservation and rollback protocol that converts fast sharing from a contention-prone setting into the strongest tested communication regime.
5. We demonstrate that the same provenance concept can be applied to a controlled cross-session LLM memory task.

The claims are limited to the tested memory rules and environments. The distance law is exact under the specified exponential gate; it is not asserted for arbitrary learned memory policies. Similarly, the LLM study demonstrates measurability and cross-session transfer, but does not constitute a broad benchmark of hallucination-resistant agent collectives.

Structure of the paper. The paper has two movements on a single object. The first (Sections 2–7) treats foreign evidence about an isolated *target*: it defines the provenance-conditioned commitment, measures its counterfactual value, derives the trust–staleness admissibility horizon, and adds the reservation-and-rollback coordination contract. The second (Sections 11 onward) shows that the transferable unit is not a target but a *relational route*: landmarks establish cross-frame identity, edges provide transport, edge-level provenance sets a rupture boundary, and only the composition reaches an unseen target. The two movements share one contract — identity, transport, provenance, validity, and coordination authority preserved together.

2 Setting

Substrate and task. N agents explore a 12×10 grid with M semantic water targets ($N > M$: scarcity) and hazards. Each agent maintains symbolic place memory (tags, confidences, support, freshness). Four sharing architectures over the identical substrate and planner: independent (no communication), shared (single event bus), centralized (consensus aggregator), peer (broadcast every K ticks, asymmetric trust), plus CSM (peer + explicit merge/trust/staleness rules with decay rate $\alpha=0.05$ and inclusion threshold $\tau=0.30$).

Q/R/M/C instrumentation. Per tick and per agent we log query formation (Q), retrieval satisfaction (R), target materialization as the planner’s lock (M^*), and completion (C^*). Episode-level conditional rates follow the companion definition. All warp measurements below are annotations of these M^* events; no planner or memory invariant is modified.

3 The warp event

Provenance and foreignness. Every evidence unit entering a merge carries its source. For the candidate m the planner locks,

$$\varphi(m) = \frac{\sum_{e \in \text{supp}(m), \pi(e) \neq \text{self}} w(e)}{\sum_{e \in \text{supp}(m)} w(e)},$$

where $w(e)$ are *the merge’s own weights*: trust $\cdot \log(1+\text{support})$ for peer/centralized consensus, trust $\cdot e^{-\alpha \text{age}} \cdot \text{conf}$ for CSM, observation masses for the shared bus. For peer and centralized architectures φ is reconstructed exactly from the per-contribution fields the merge already exports; the measurement layer reads state and never writes it.

Freezing rule. φ is fixed at the moment of the lock. If the agent later confirms the place with its own eyes, φ is *not* recomputed — otherwise every successful warp would dilute itself on approach and $P(C^*|W^*)$ would be unmeasurable by construction.

Warp events and strata. $W_\theta^* := M^* \wedge (\varphi \geq \theta)$ with $\theta=1$ (strict: never self-observed, cross-checked against an observation ledger) and $\theta=0.8$ (soft). Warp share $P(W^*|M^*)$ is a structural descriptor (identically 0 for independent); warp completion $P(C^*|W^*)$ is the new quantity. Under the Q/R/M/C event contract defined in the companion evaluation study, warp is an exact stratification of the materialization event M^* rather than an additional stage: identifiability of Q/R/M/C is untouched.

Regression anchor. The known misled-agent episode of the architecture comparison (agent A locks (3, 8) known only from C; C claims it first; A strands) is annotated by the instrumentation as $\varphi = 1.0$ strict W^* with collision — the pathology the framework must see, and the protocol of §6 must fix.

4 Direct and downstream effects of foreign evidence

Design. 2,400 episodes: $(N, M) \in \{(3, 2), (5, 3), (8, 5), (4, 4), (6, 2)\} \times \text{three layouts} \times \{\text{peer-}K2, \text{peer-}K8, \text{CSM-}K8, \text{shared}\} \times 20 \text{ seeds} \times \{\text{full, foreign-masked}\}$ with identical seeds. Unit of analysis: lock event; CIs by episode-cluster bootstrap.

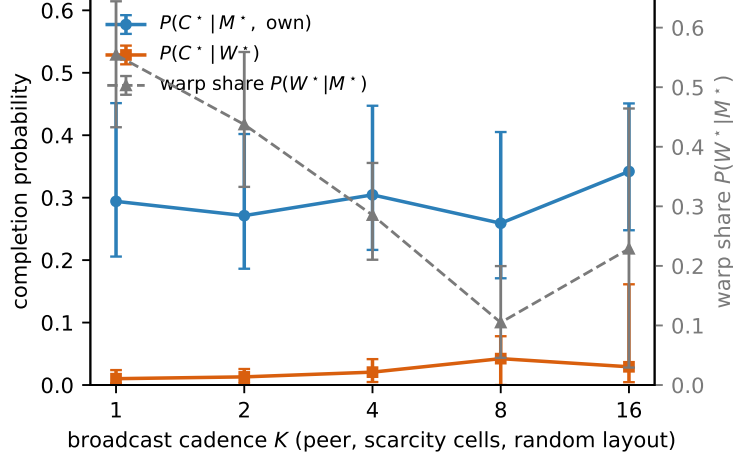


Figure 1: φ -stratified completion vs. broadcast cadence K (dedicated peer sweep: scarcity cells, random layouts, 20 seeds, episode-cluster bootstrap CIs). As K shrinks, the warp share $P(W^*|M^*)$ grows from 0.11 to 0.55 while $P(C^*|W^*)$ stays below 0.05 and the own-stratum rate stays flat: the completion collapse of fast sharing is concentrated in the W^* stratum, giving the Q/R/M/C bottleneck shift reported in the companion evaluation study its mechanistic reading. The share uptick at $K=16$ (broadcasts are rare, but agents also discover little on their own) is reported as observed.

Figure 1 shows the cadence view on a dedicated peer sweep ($K \in \{1, 2, 4, 8, 16\}$): warp share is what fast sharing buys, and it is precisely the stratum that does not complete. Table 1 gives the architecture view.

Table 1: Provenance strata of completion under scarcity ($N > M$), full runs. Warp locks complete an order of magnitude less often than own-evidence locks; all CI pairs are disjoint. (H-W1 confirmed.)

Architecture	$P(C^* W_{0.8}^*)$	n_{W^*}	$P(C^* M^*, \text{own})$	n_{own}
peer- $K=2$	0.004 [0.001, 0.008]	1,437	0.223 [0.186, 0.275]	2,738
peer- $K=8$	0.020 [0.000, 0.038]	149	0.389 [0.329, 0.446]	1,734
CSM- $K=8$	0.017 [0.006, 0.044]	637	0.414 [0.368, 0.453]	1,684
shared	0.004 [0.002, 0.008]	2,099	0.279 [0.237, 0.332]	2,250

4.1 Downstream transport through lock chains

Direct completion understates the value of foreign evidence. A *warp-assisted own completion* is a completed own-evidence lock preceded (same agent, same episode) by a soft- W^* lock that was dropped: the warp failed as a destination but succeeded as a ride.

Table 2: Lock-chain analysis (random layouts, full runs). The first W^* lock in a chain is made from ≈ 8 –10 cells away; the completing own lock from 2.0 cells — exactly the observation radius, i.e. the self-confirmation point.

Arm	own C^*	warp-assisted (share)	$r(\text{first } W^*) \rightarrow r(\text{own})$
shared	248	127 (51.2%)	9.5 $\rightarrow$ 2.0
peer- $K=2$	223	99 (44.4%)	9.7 $\rightarrow$ 2.0
CSM- $K=8$	279	13 (4.7%)	5.8 $\rightarrow$ 2.0
peer- $K=8$	246	7 (2.8%)	8.4 $\rightarrow$ 2.0

This resolves an apparent paradox in the counterfactual data: the shared bus at moderate scarcity gains +10.8 ticks [+2.2, +20.9] from foreign evidence although its direct $P(C^*|W^*)$ is 0.004. The gain flows through chains; the freezing rule books the completion in the own-evidence stratum, and the chain analysis recovers the warp’s contribution as a separate measured quantity.

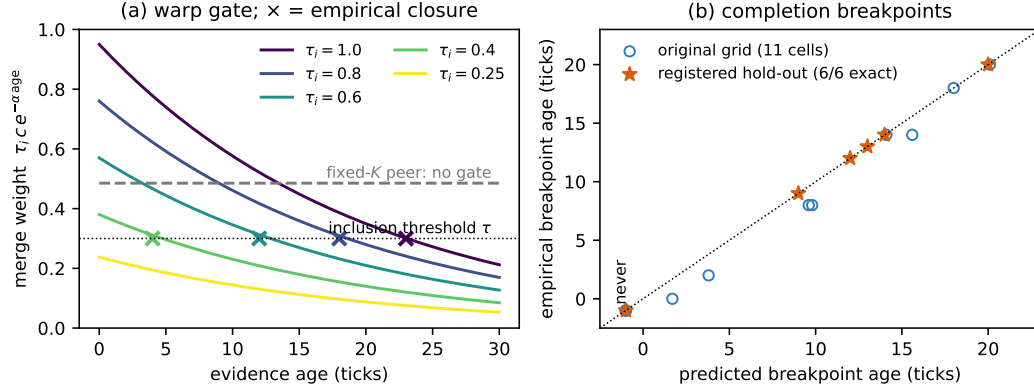


Figure 2: The warp distance law. **(a)** Foreign-evidence merge weight under trust $\times$ staleness decay; $\times$ marks the empirically measured last age at which the traveller’s query still exposes the target — each sits on its curve’s crossing of the inclusion threshold τ . Fixed-cadence peer memory has no staleness term (dashed): its gate never closes; the architectures differ in the *shape* of the curve. **(b)** Predicted vs. empirical completion breakpoints in full navigation episodes: original grid (circles; age sampled in steps of 2, hence points at most one grid step below the diagonal) and the registered hold-out (stars; age step 1, predictions written to disk before execution, 6/6 exact, including the never-cell at $(-1, -1)$ and the cell where the pruning horizon undercuts the exponential gate).

4.2 Counterfactual warp gain

Masking foreign evidence at merge under the same seed gives $WG = t_{\text{cens}}(\text{masked}) - t_{\text{cens}}(\text{full})$ per episode. The sign map over (layout, scarcity $\rho = N/M$) is regime-dependent: positive for the shared bus at unpredictable layouts and moderate scarcity ($+10.8$ at $\rho = 1.5$), negative for fast peer sharing (peer- $K2$ random: $-7.1 [-13.5, -1.3]$), ≈ 0 at hard scarcity. A per-agent mask (one agent masked, others share normally) confirms the sign at the individual level: $WG_i = -10.25 [-20.91, -0.16]$ ticks on the fast-share scarcity cell, with spillover on unmasked agents ≈ 0 ($-0.62 [-3.06, +1.91]$). Raw warp *hurts* its recipient precisely where sharing is fastest — the collision cost exceeds the information value.

Exploratory: collision pressure. Co-lock pressure (other agents locked on the same target) correlates with warp failure with the predicted sign but negligible magnitude (Spearman $+0.033$, $p = 0.018$, $n = 5,117$); we report it as exploratory rather than as a confirmed predictor.

5 The warp distance law

CSM includes foreign evidence with weight $\tau_i e^{-\alpha \text{age}} c$ (trust τ_i , confidence $c = 0.95$) and inclusion threshold $\tau = 0.30$. Foreign evidence therefore stops forming intention at

$$\text{age}_{\max}(\tau_i) = \frac{1}{\alpha} \ln(\tau_i c / \tau), \quad \text{defined for } \tau_i c > \tau.$$

Witness-traveller protocol. A witness observes water once and goes silent; a traveller who has never seen it receives the snapshot at a parameterised age. Every lock the traveller makes is a strict W^* by construction. Figure 2 summarises both experiments.

Completion breakpoint. In full navigation episodes the feasibility inequality acquires one discrete refinement: the traveller self-confirms the target once it is within the observation radius, so the foreign gate only has to survive until tick $t_{\text{gate}} = d - r_{\text{obs}} - 1$, giving breakpoint $\text{bp} = \text{age}_{\max} - t_{\text{gate}}$; snapshot pruning executes on broadcast ticks and adds the bound $10K - t_{\text{bcast}}$. All constants of the law are unchanged. All 11 CSM cells of the original grid match within sampling resolution; completion curves are monotone steps; the trust-flip zeroes W^* events discretely.

The law therefore predicts the held-out horizons exactly at tick resolution: the CSM rules really are the mechanism that forms (and un-forms) intention. It also reinterprets the companion result that

Table 3: Gate curve: largest evidence age at which the traveller’s query still exposes the target. Fixed-cadence peer memory has no staleness term: the gate never closes (checked to age 120 at all trusts) — the architectures differ in the *shape* of the curve, not just its level.

trust τ_i	empirical age _{max}	predicted
1.00	23	23.05
0.80	18	18.59
0.60	12	12.84
0.40	4	4.73
0.25	gate closed	0 (trust-flip)

Table 4: Registered-prediction hold-out: six cells absent from the original grid, predictions written to disk *before* any episode, age grid step 1. All six are exact integer matches, including the never-cell and the cell where the pruning horizon must undercut the exponential gate.

Cell	binding bound	predicted	empirical
$K=8, \tau_i=0.9, d=9$	gate	14	14
$K=8, \tau_i=0.7, d=9$	gate	9	9
$K=8, \tau_i=0.5, d=15$	gate	never	never
$K=4, \tau_i=1.0, d=6$	gate	20	20
$K=2, \tau_i=1.0, d=12$	<i>pruning</i>	12	12
$K=2, \tau_i=0.8, d=8$	gate	13	13

CSM dominates fixed-cadence peers: trust $\times$ staleness is a *warp gate* — CSM does not share better, it *refuses to warp on a stale or untrusted song*.

Portability to continuous dynamics. All three registered predictions survive on a continuous VMAS substrate through the same continuous $\rightarrow$ grid bridge used by the companion’s portability test: (i) the provenance strata stay an order of magnitude apart ($P(C^*|W^*) = 0.012$ [0.000, 0.042] vs. 0.562 [0.457, 0.709] at $K=1$); (ii) warp share falls monotonically with cadence ($0.723 \rightarrow 0.366 \rightarrow 0.000$ over $K=1/4/16$); (iii) the law becomes *metric*: with the travel-to-sight time calibrated once on a fresh-evidence episode, the predicted breakpoints match within the sampling grid, and all cells beyond the metric warp radius ($v \cdot \text{age}_{\text{max}}$ shorter than the distance) are predicted *never* and observed *never*. One genuinely continuous finding: inertia is a physical warp-leak channel — an agent can coast into sighting range after the gate closes — removed by a no-lock-then-brake controller that restores the grid semantics.

6 The Warp Drive protocol

Warp collision — locking on foreign evidence for a target someone else will claim — is the dominant failure of fast sharing. Warp Drive adds three primitives, all decentralized:

1. **Optimistic warp lock** — locks on foreign evidence remain allowed immediately (planner unchanged).
2. **Warp reservation** — at the owner’s next broadcast tick the reservation rides on the ordinary snapshot (same cadence K , no coordinator, no extra channel); receivers suppress reserved targets in their queries. Priority: (lock tick, distance at lock, agent id).
3. **Anti- M^* rollback** (Time Warp semantics Jefferson (1985)) — an agent holding a lock dominated by a foreign reservation retracts it: the planner re-queries, the target enters exponential backoff ($4 \rightarrow 8 \rightarrow \dots \rightarrow 32$ ticks) with an agent-id tie-break against oscillation. The event log stays append-only.

Decentralisation check. Driving the protocol by direct per-agent calls — no runtime, no environment — produces the same retraction decisions from each agent’s own inbox alone.

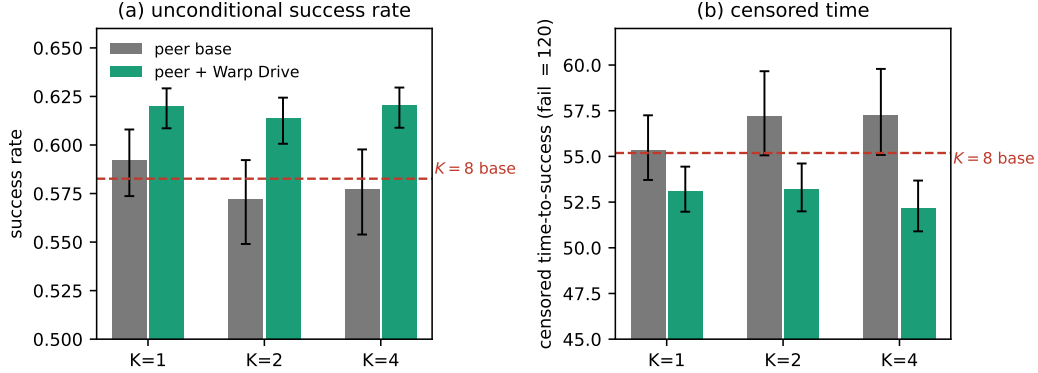


Figure 3: Warp Drive on unconditional metrics (840 episodes; bootstrap CIs). **(a)** Success rate rises at every cadence with disjoint CIs, and every fast- K protocol arm exceeds the $K=8$ sweet-spot baseline (dashed). **(b)** Censored time-to-success falls correspondingly. The claim that the protocol removes the main argument against fast sharing rests on these unconditional quantities, not on the (denominator-sensitive) conditional rates.

Misled-A anchor. With Warp Drive the pathological lock is resolved *preventively*: C’s reservation arrives before A ever locks (3, 8); A completes on another target. In the sweep, rollback latency of dominated locks is finite and $\approx K$: 1.0/1.5/2.7 ticks at $K=1/2/4$ (baseline: until end of episode).

Table 5: Unconditional results, 840 episodes (scarcity cells $\times$ random/asymmetric layouts $\times$ 20 seeds). Success rate rises under Warp Drive at every cadence with disjoint CIs, censored time falls, and fast sharing with the protocol beats the $K=8$ sweet-spot baseline (Figure 3). Lock census shows the conditional-metric denominator shrinking — which is why the claim is stated on unconditional quantities.

Arm	success rate	t_{cens} (fail = 120)	lock census
$K=1$ base	0.592 [0.574, 0.608]	55.3 [53.7, 57.2]	0.986
$K=1$ +WD	0.620 [0.609, 0.629]	53.1 [52.0, 54.4]	0.985
$K=2$ base	0.572 [0.549, 0.592]	57.2 [55.1, 59.7]	0.973
$K=2$ +WD	0.614 [0.601, 0.624]	53.2 [52.0, 54.6]	0.926
$K=4$ base	0.577 [0.554, 0.598]	57.3 [55.1, 59.8]	0.944
$K=4$ +WD	0.620 [0.609, 0.630]	52.2 [50.9, 53.7]	0.739
$K=8$ base	0.583 [0.558, 0.604]	55.2 [52.9, 57.9]	0.682

On the conditional axis, recovery is measured against the scarcity ceiling $\min(m, M)/m$ (no protocol conjures extra water): 0.90/2.15/6.6 of the recoverable collapse at $K=1/2/4$; values above 1 arise because reservation suppression removes doomed locks from the M^* population. Together with Table 2, this completes the arc: raw warp fails at completion (§4); with a coordination contract attached, warp at fast cadence becomes the best regime. The cadence sweet spot of the companion paper is reinterpreted: moderate K was never about information freshness — it was an implicit rate limiter on warp collisions, and an explicit contract does the job better.

7 Cross-session warp on an LLM substrate

Finally, the strict warp crosses both an agent boundary and a session boundary on a language substrate. In session 1, agent A (a local LLM tag extractor over natural-language observations; Ollama llama3.1, deterministic caching, zero API cost) sweeps a three-room text household and its symbolic memory is dumped to disk. In session 2, agent B — a fresh agent in a fresh environment instance, with the stock directional oracle removed so that long-range knowledge can only come from memory — loads A’s dump as foreign provenance and solves “bring the apple to the table.” Every lock B makes on the apple or the table is $\varphi=1.0$ strict W^* ; execution is layered (the LLM makes the

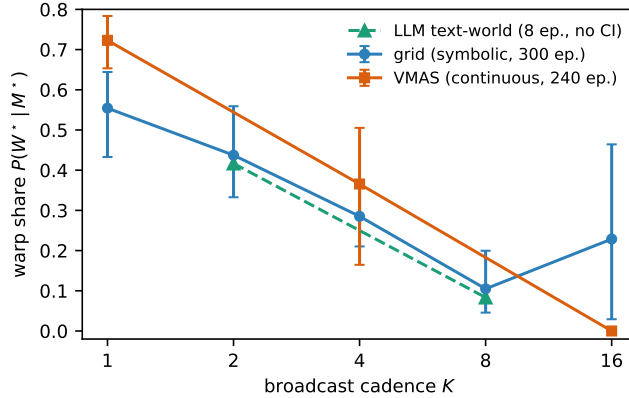


Figure 4: One quantity, three substrates: warp share $P(W^*|M^*)$ vs. broadcast cadence K on the symbolic grid, the continuous VMAS scenario, and the LLM text-world. The curves nearly coincide — how much of a collective’s materialization runs on foreign evidence is set by the sharing cadence, not by the substrate. (LLM arm: 8 episodes, point estimates only.)

semantic decisions — which remembered place to pursue, when to take/put — while locomotion to the locked waypoint is a deterministic motor primitive, matching the substrate’s planner stack).

Table 6: Cross-session warp over 10 randomised layouts. Completion is attributed at the environment’s affordance radius (take/put act within Manhattan distance 1, and the witness tags anchors on within-reach cells).

	success	mean t_{succ}	layouts with completed strict W^*
warp (A’s memory)	10/10	8.3	10/10
control (empty memory)	0/10	—	—

This experiment provides a controlled demonstration of cross-session, cross-agent strict semantic warp on an LLM substrate. The 10/10 versus 0/10 contrast shows that foreign memory is necessary for completion in this controlled task.

Scaling to a live collective. With $N=3$ LLM agents sharing by peer broadcast under scarcity ($M=2$ apples, four layouts, $K \in \{2, 8\}$), the full warp phenomenology reproduces on the language substrate (all predictions registered before the runs): strict W^* locks occur; warp share falls with cadence (0.417 at $K=2$ vs. 0.083 at $K=8$ — the same shape as the grid $0.554 \rightarrow 0.105$ and VMAS $0.723 \rightarrow 0.000$); warp collisions appear in 7/8 episodes. A paired comparison against the no-communication baseline on identical layouts shows raw sharing *lowering* success from 2/3 to 1/3 per episode — the collision cost exceeds the information value on the LLM substrate exactly as in §4.2. Attaching the unchanged Warp Drive protocol to the LLM collective provides a corresponding language-agent evaluation: pooled success rises from 0.333 to 0.417 at both cadences with rollback latencies again $\approx K$ (1 tick at $K=2$, 7 at $K=8$). The recovery is partial ($\approx 25\%$ of the collapse, vs. 90–100% on the grid): on a language substrate the collision pathology is compounded by semantic-level losses (hallucinated anchor tags, noisy locks) that no reservation can cure — a measurable decomposition we leave to future work. The distance law also survives LLM-produced evidence: with the gate constant c set to the confidence the extractor actually emitted ($c_{\text{meas}}=0.900$), all four registered breakpoints land within one tick of prediction.

Two interface findings transfer to any LLM-over-symbolic-memory design: the LLM query former free-generates retrieval tags outside the memory’s vocabulary (requiring an any-of gate), and small local models are reliable at semantic commitment but not at coordinate arithmetic or systematic exploration (requiring the layered execution above and an explicit exploration tier — without one, discovery never happens and the warp has nothing to feed on).

Figure 4 overlays the warp-share–cadence curve across all three substrates: the quantity behaves as an architecture-level pattern that remains qualitatively consistent across the three tested substrates.

8 From target commitments to relational routes

This first movement introduced semantic warp as a provenance-conditioned annotation of a memory-based commitment. The annotation makes it possible to distinguish direct observation from peer-supported action, to estimate the counterfactual contribution of foreign memory, and to derive an explicit temporal boundary for its authority under a trust–staleness gate.

The experiments show that foreign evidence should not be evaluated only by the direct completion rate of foreign locks. Such locks are fragile under scarce targets, but they can provide substantial downstream value by transporting recipients into self-confirmation range. The same analysis localizes the principal cost of fast sharing to duplicated commitments rather than to evidence quality. A minimal reservation-and-rollback protocol addresses this failure and improves unconditional success across the tested fast cadences.

The resulting conclusion is narrower than a general claim that shared memory is beneficial. Foreign evidence becomes useful when its source, age, and authority are represented explicitly and when its use is coupled to a coordination protocol. The remaining sections extend this result from isolated target commitments to provenance-bearing route structures and remove the shared-frame assumption.

Outlook: the contract as one function among several. The contract perspective suggests a reading we state as a falsifiable program rather than a conclusion: memory is not one mechanism but several functions with distinct failure signatures. The trust×staleness gate studied here implements *memory-for-trust* (ablate it and completion collapses through stale and poisoned commits — the trust-flip and the raw-exchange negative are exactly that signature); it is distinct from *memory-for-return* (own consolidation; its ablation starves candidate generation) and from *memory-for-telling* (what is worth broadcasting at all; its ablation leaves receivers’ retrieval unimproved on tags they undersample). Under this reading the right validity criterion for shared memory is not direct verification of content but *ecological viability*: whether a commitment formed on foreign evidence can still be carried to completion under the contract — which is precisely the quantity $P(C^* \mid W^*)$ and its law make measurable.

9 Routes and meaning-based identity: the transferable unit

What transfers between agents is the song: edge-provenanced routes, landmark-anchored place identity without shared frames, the vocabulary ablation, and the song anatomy.

Transferring the location of a target does not necessarily transfer an executable navigation strategy. In environments with obstacles, hazards, or private coordinate frames, a recipient may know where a target is believed to be while lacking both the route required to reach it and a reliable correspondence between the sender’s places and its own observations. We therefore model the transferable memory unit as a provenance-bearing relational route. Semantic landmarks identify corresponding places, route edges encode transport between them, and edge-level provenance records which parts of the route are known only through another agent. A strict route-warp event is defined as the execution of a connected foreign subpath that the recipient has not previously traversed. Experiments show a clear transition between place-only and route-level transfer: isolated target records fail at the first obstructing structure and can perform worse than unguided exploration, whereas transferred routes remain executable. Because route edges are observed at different times, their trust-weighted evidence decays non-uniformly. The resulting rupture law predicts the first inadmissible edge in all 110 tested regimes and with zero index error on 21 intermediate ruptures. Semantic landmark constellations recover place correspondence across translations and quarter-turn rotations, while ambiguous configurations are rejected rather than materialized as phantom targets. A representation ablation further separates the two informational functions of the route: landmarks provide identity and edges provide transport; only their composition reaches an unseen target. In the tested codec, this composition uses 6.4% of the bits of a full snapshot before protocol metadata. The results support relational, provenance-conditioned route transfer as the appropriate unit for executable foreign memory. The study remains limited to controlled semantic observations and discrete frame transformations; continuous pose estimation and open-ended perceptual landmark discovery are outside its scope.

9.1 From target records to executable routes

Collective navigation memory is commonly represented as a set of places, coordinates, or target descriptors. Such representations can support target retrieval, but they omit two requirements of executable transfer. First, the recipient needs a transport structure that specifies how the relevant places are connected. Second, the sender and recipient must determine whether their local records refer to the same physical places, even when their coordinate frames differ. A shared target label or a nearby coordinate is insufficient when either of these conditions is absent.

This section addresses both requirements by defining a song as a relational route with semantic landmarks, ordered transitions, and edge-level provenance. The term is used as a technical representation: landmarks provide candidate correspondences between local memories, whereas route edges specify the sequence of transitions that makes a target operationally reachable. Provenance indicates which route segments were acquired by the sender and remain unverified by the recipient.

The first question is whether route structure provides measurable value beyond place-only transfer. We evaluate this through controlled mazes in which the target coordinate is correct but the direct path is blocked. The second question concerns temporal validity. Because the observations supporting different route edges have different ages, the transferred path can remain admissible only up to a finite prefix. We derive a rupture index by replaying the trust–staleness gate along the route. The third question is referential: can agents recover the same route when their private frames differ? We use semantic constellations around landmarks to generate mutually consistent correspondences and reject ambiguous matches.

The route-level analysis contributes the following:

1. We extend provenance from target commitments to route edges and define strict route warp as the execution of a connected foreign subpath not previously traversed by the recipient.
2. We quantify the performance difference between place-only and route-level transfer and identify the structural conditions under which target coordinates are insufficient.
3. We derive a route-rupture boundary from edge-specific trust–staleness values and validate its predicted failure edge across the registered test regimes.
4. We introduce meaning-based place correspondence through semantic landmark constellations and evaluate frame recovery and fail-closed behavior under ambiguity.
5. We decompose the route representation into identity-bearing landmarks and transport-bearing edges and measure the storage cost of their composition relative to a full snapshot.

The central claim is therefore not that relational or topological navigation is new in itself. The contribution is the composition of route structure, provenance, temporal validity, and semantic place correspondence into a measurable transfer object. The present experiments use controlled semantic vocabularies and discrete translations and quarter-turn rotations. Arbitrary continuous $SE(2)$ estimation, rich visual observations, and learned landmark proposal remain open.

10 Route-transfer setting

Grid worlds (12×10 to 14×12) with semantic water targets, hazards, and walls; agents with symbolic place memory, peer broadcast on cadence K , and the common memory-grounded planner used in all experiments (target lock = the Q/R/M/C M^* event; completion C^*). The trust $\times$ staleness inclusion rule defined in the companion collective-memory study is used throughout with unchanged constants: evidence mass $\text{trust} \cdot e^{-\alpha \text{age}} \cdot m$, threshold $\tau=0.30$, $\alpha=0.05$. Warp events follow the provenance analysis of the first part of this paper: provenance φ frozen at lock time; completion attributed to a continuous lock from a foreign commit ($\varphi=1$ at first lock) to success.

11 Route-level transfer

Edge provenance. Broadcast snapshots additionally carry the sender’s OWN traversed edges ($a, b, \text{count}, \text{last tick}$); foreign edges are never re-broadcast (the same privacy contract as place

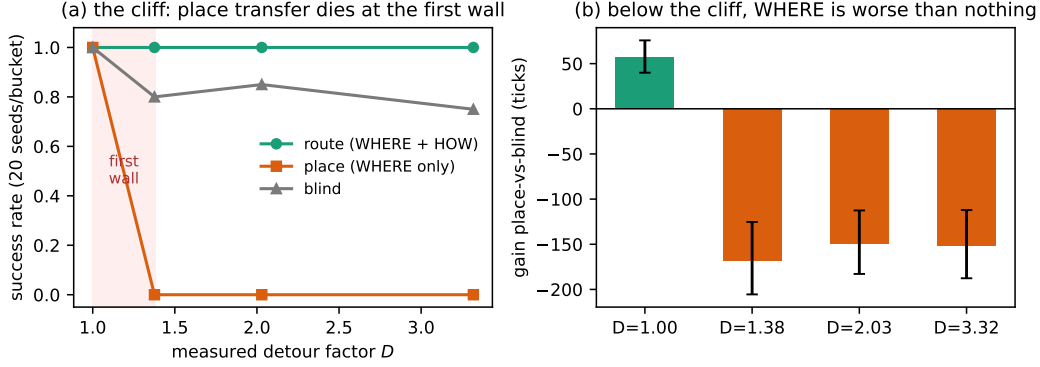


Figure 5: The cliff (paired arms on identical mazes, 20 seeds per measured-detour bucket). **(a)** Place-only transfer does not degrade with detour — it collapses at the FIRST wall, located by the probe bucket between $D=1.00$ and $D=1.38$; route transfer completes everywhere. **(b)** Below the cliff, knowing WHERE without HOW is *worse than knowing nothing*: the known-target manhattan pull pins the planner against walls (−150 to −168 ticks vs. blind exploration), while in the open world the same knowledge is worth +57.

snapshots: no agent reads another’s graph, transport only). For the path P a planner commits,

$$\varphi_{\text{route}}(P) = \frac{\sum_{e \in P, \pi(e) \neq \text{self}} w(e)}{\sum_{e \in P} w(e)}, \quad w(e) = \text{trust} \cdot e^{-\alpha \text{age}(e)} \cdot \log(1 + \text{count}(e)),$$

the same weights the route gate uses. A **strict RW^*** is a committed, traversed, connected sub-path of ≥ 5 edges with $\varphi_{\text{route}} = 1$: the whole verse is someone else’s. Place warp is the degenerate case (path length 0: only the target is foreign), giving a two-dimensional provenance plane $(\varphi_{\text{place}}, \varphi_{\text{route}})$.

First strict RW^* . On a comb maze with measured detour $D=2.65$, a traveler commits and walks a fully foreign 45-edge route to completion in 54 ticks; with the same foreign PLACE evidence but no edges, the standard planner never arrives (300-tick budget), and blind exploration takes 126.

11.1 Performance transition under route transfer

Table 7: Three arms on identical mazes (paired by seed), bucketed by MEASURED detour factor D ; 20 seeds per bucket, censored gains (fail = 300). Place-only transfer does not degrade with D — it collapses at the first wall, and is worse than blind exploration there (the known-target manhattan pull pins the planner against walls). Route transfer completes everywhere; every completion is a strict RW^* with $\varphi_{\text{route}} = 1$.

D	gain route-vs-place [CI]	gain route-vs-blind [CI]	succ r/p/b
1.00	+4.7 [+3.4, +6.0]	+62 [+44, +80]	20/20/20
1.38	+281 [+279, +283]	+113 [+77, +155]	20/0/16
2.03	+273 [+271, +275]	+123 [+91, +160]	20/0/17
3.32	+262 [+259, +264]	+110 [+72, +152]	20/0/15

Figure 5 shows both faces of the result. Two pre-registered predictions failed here, in instructive ways. The negative control at $D=1$ was registered in ticks ($|\text{gain}| \leq 3$) and came out +4.7: the gap is *turn economy* (BFS routes walk straight legs; the dominant-axis planner alternates axes), not route knowledge — in path LENGTH the gap is exactly 0.0 [0.0, 0.0] across all 20 open-grid cells. And “gain grows with D ” is unmeasurable: with place success at 0 the gain saturates at the censoring ceiling. The correct form of the law is a step located by the probe bucket at $D \in (1.00, 1.38)$, i.e. the first wall.

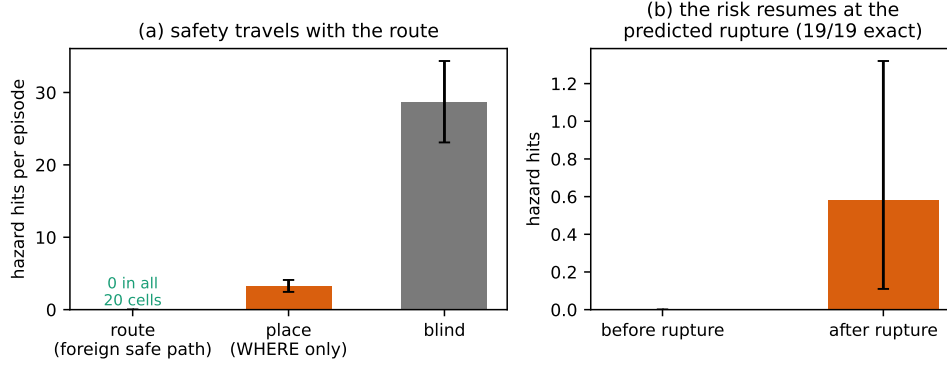


Figure 6: The song protects. **(a)** Hazard hits per episode: the traveler on the witness’s route preserves the witness’s measured risk profile within the validated route prefix (0 in all 20 worlds); the place-only traveler pays 3.25 hits for the same completion; blind exploration pays 28.7. **(b)** Composition with the rupture law: in the rupture regime the traveler takes zero hits while the song lasts, and the risk resumes after the break — which occurs at the predicted rupture index in all 19/19 cells.

12 Rupture law and route-validity boundary

The witness stamps edge i at tick i : its trace ages non-uniformly, the start of the route always older than the end, and a traveler delayed by a_0 *chases a fading song*. Under the standard gate the binding edge is the CURRENT one (the oldest remaining), giving three regimes on the a_0 axis: complete; rupture at a predictable path index; dead on arrival. The predictor is fully deterministic — a no-decay calibration run records the traveler’s kinematics (path index per tick, turns included), and the registered prediction replays the closed-form gate over that trajectory, the same deterministic replay procedure used for the 6/6 hold-out in Section 5.

Across 5 mazes $\times$ 11 delays $\times$ 2 trusts (110 cells, registered before execution): **110/110 regime matches; all 21 mid-route ruptures at exactly the predicted edge index (error 0)**; and a route trust-flip — at trust 0.7 the single-traversal edge mass gives $\text{age}_{\max} = 9.6 < L$, so *no route ever commits* (0 commits observed; the discrete off-switch of the warp distance law of Section 5, now for paths). The rupture of a transferred route is thus not noise but a law: the same constants (α, τ , trust) that set WHEN a place-warp gate closes set WHERE a route-warp breaks.

13 Hazard preservation along the valid route prefix

Table 8: Open fields dense with passable-but-punished hazards (density 0.20), the witness’s route hazard-avoiding by experience; 20 paired worlds, decay off (the safety axis isolated from the aging axis). Both transfer modes COMPLETE — the wall axis (Table 7) and the hazard axis are independent; they differ in the price.

Arm	hazard hits [CI]	completion	mean t
route (foreign safe path)	0 (all 20 cells)	20/20	16.6
place (WHERE only)	3.25 [2.45, 4.10]	20/20	15.9
blind	28.7	—	—

The traveler following the witness’s path preserves the witness’s measured risk profile within the validated route prefix, at a time premium of +0.7 ticks (Figure 6a). Composing with §12: in the rupture regime (19/20 worlds) the traveler takes **0 hits before the rupture** and 0.58 [0.11, 1.32] after it, when it falls back on place knowledge and re-enters the field — with all 19 ruptures again at the predicted rupture index (error 0; the edge-exact predictor replicated on a new world class with no adjustment). The song protects while it lasts, and the measured protection ends at the edge predicted by the rupture law.

14 Meaning-based place correspondence

Everything so far in this paper rides on a shared coordinate frame. We now remove it: each agent’s private frame is the true grid under a secret translation (later: plus a secret 90° -multiple rotation), and broadcasts carry private coordinates that are meaningless to the receiver until it decides what the sender’s places MEAN.

Two levels of ‘the same’, and what a landmark is. Cross-agent place identity has two levels that must not be conflated. The *semantic predicate* says two cells belong to the same desired class (water_source, GOOD): it travels in every snapshot, and it is what task satisfaction is defined over. But no predicate can decide *entity correspondence* — that two memory entries denote the same world referent — because a predicate is exactly what aliases: all water looks like water. Route-relative description (‘ n steps past the last GOOD’) does not decide it either: it is provenance of a path, not identity of its endpoints (§11–§12 measure what it buys and where it breaks). What decides correspondence in this series is a *landmark*: a place whose local semantic constellation supports a mutually-unique match across agents. Landmark is a *role*, not a cell type — which tag classes can play it is itself measured below (§14.2).

Fingerprints and the handshake. A place fingerprint is the local semantic constellation around a visited cell — which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Matching is *mutually unique* (a pair is accepted only if each side is the other’s sole candidate above an exact-equality threshold; the sender knows far more places than the receiver, and one-directional uniqueness aliases distant look-alikes into the receiver’s neighborhood), fingerprints need ≥ 2 salient keys, and the frame translation is the modal delta of ≥ 3 agreeing pairs. Landmarks must carry at least one CONTENT tag: border-only (‘void’) patterns describe the shape of the world, not a place, and are excluded — the lesson of the corner trap below.

Witness–traveler results. Over $4 \text{ offsets} \times 2 \text{ hazard densities} \times 10 \text{ seeds}$: the semantic aligner recovers the secret offset *exactly in 100%* of landmark-world runs (mean 15.1 mutual pairs) and completes with strict W^* ($\varphi=1$) every time; the coordinate baseline under any nonzero offset locks a non-water cell in 100% of runs (its rare ‘successes’, 6.7%, are the displaced route accidentally crossing the true water). In the aliased world (featureless interior band) the two identity regimes break in OPPOSITE directions: semantic identity makes *zero* locks — it fails closed — while coordinate identity keeps confidently locking wrong cells — it fails open. This asymmetry, not raw accuracy, is the deepest argument for meaning-based identity: when the world does not support identification, the semantic system knows it.

14.1 Integrated correspondence and route-transfer stack

Online alignment in the peer stack. With $N=4$ agents under scarcity, all in distinct secret frames, alignment develops DURING the episode: agents explore, co-visited landmarks accumulate, and per-sender frames lock in mid-run (onset in 100% of episodes, mean tick ≈ 15). Pooled success: shared-frame oracle 0.456, coordinate under misalignment 0.350, semantic 0.444 — **88% of the gap recovered**, with strict W^* functioning through recovered frames.

Validity law under recovered frames. Re-running the warp distance law of Section 5 with the trust $\times$ staleness gate ON TOP of semantic identity under a frame offset gives breakpoints 20/14/9 — *the same integers as the shared-frame closed form, exactly*. The gate and the frame recovery are orthogonal layers: the law does not notice that the shared grid is gone.

No shared north: translations and quarter-turn rotations. With secret rotations added, one rotation hypothesis per 90° reuses the translation aligner and the winner must *dominate* ($\geq 2\times$ the runner-up’s consensus). Results: exact (r^*, δ^*) recovery in **80/80** rotation $\times$ offset $\times$ seed combinations with 100% strict- W^* completion; in a 180° -symmetric world — where the frame is unrecoverable *in principle* — the aligner refuses (0/10 locks) while coordinates fail open (10/10 wrong locks); and the distance law again does not notice (bp = 20 under a 90° -rotated witness). The translation-only aligner, for contrast, never recovers a rotated frame (0/60) but does not always fail closed: in 2/60 cells a 180° -symmetric hazard substructure lets it fail OPEN — the empirical motivation for the dominance rule.

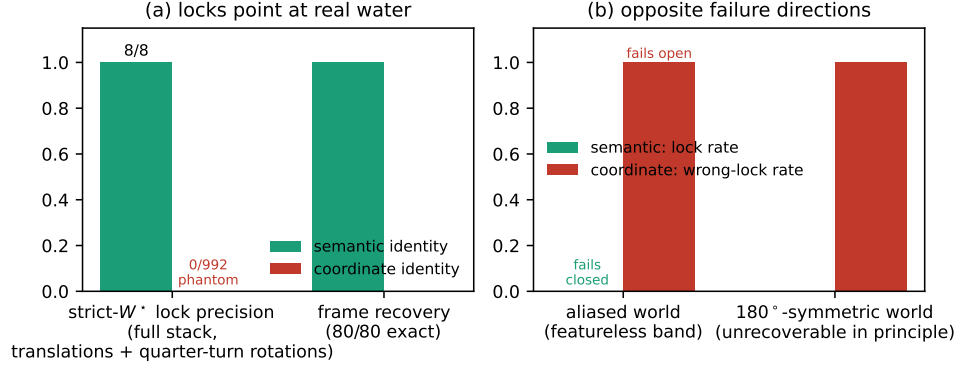


Figure 7: Meaning-based identity under discrete rigid-frame transformations ($\mathbb{Z}^2 \rtimes C_4$: translations and quarter-turn rotations). **(a)** Every strict- W^* lock the semantic stack makes points at real water (8/8); the coordinate stack makes 992 locks, all phantoms (0/992); frame recovery itself is exact in 80/80 witness–traveler combinations. **(b)** The failure asymmetry in worlds where identification is impossible (featureless band; a 180°-symmetric layout where the frame is unrecoverable in principle): semantic identity refuses to lock — fails closed — while coordinate identity keeps confidently locking wrong cells — fails open.

The corner trap. In the full stack, raw success is non-discriminating under rotations (rotated coordinate poison mostly maps off-grid and degrades into pseudo-exploration), so the registered discriminator is *warp-lock precision*: the share of strict W^* locks pointing at real water. Before the void filter, early border fingerprints (agents start in corners; rectangle borders are 180°-symmetric) fed wrong-frame hypotheses and semantic precision was 0.159; with borders excluded as non-landmarks it is 1.000 (8/8) **vs.** 0.000 (0/992 **phantom locks**) **for coordinates**, with alignment onset in 100% of episodes and clean regressions on all earlier experiments (Figure 7).

14.2 Causal contribution of the landmark vocabulary

The identity machinery rides on a fixed landmark vocabulary $\Sigma = \{\text{wall, hazard, water, void}\}$: the tag classes allowed into a fingerprint. Every claim above is therefore conditional on a hand-chosen alphabet. The last experiment makes the alphabet the treatment: witness–traveler frame recovery (worlds carrying both scattered walls and scattered hazards; 3 offsets $\times$ 12 seeds per arm) under seven vocabularies, from full Σ down to single classes.

Table 9: Landmark-vocabulary ablation (36 cells per arm, 252 total; registered predictions A1–A4, all PASS). Impoverishing the alphabet costs *coverage* — recovery converts into refusal — but never *correctness*: every transported lock, under every vocabulary, goes through the exactly-recovered frame. Attribution = share of consensus-pair key mass under full Σ .

Vocabulary	exact recovery	mean pairs	lock precision	fail-open
full Σ	1.00	33.2	1.00	0
– void	1.00	32.0	1.00	0
– hazards	0.92	11.8	1.00	0
– walls	1.00	15.5	1.00	0
hazards only	1.00	13.7	1.00	0
walls only	0.83	10.8	1.00	0
void only	0.00	0.0	—	0

Four registered predictions, four passes (Table 9). **(A1) The ablation is fail-safe:** in 252 cells there is not one fail-open transport, consistent with the fail-closed asymmetry of §14 holds not just against ambiguous *worlds* but against impoverished *alphabets*. **(A2) Void is scaffolding, not identity:** border patterns carry 0.000 of the consensus key mass, and dropping them changes nothing — consistent with the corner trap, where they actively poisoned rotation hypotheses. **(A3) One content class suffices:** hazards alone recover every frame; walls alone recover 0.83 and refuse the rest. Correspondence

needs only ≥ 3 mutually-unique constellations over *some* content class — a bounded-rationality budget, not a rich ontology. **(A4) Attribution is measurable:** hazard constellations carry 0.56 of the consensus mass, walls 0.44. Which features function as landmarks is thus not a modelling assumption that happened to work — it is a measurable property of the world $\times$ vocabulary pair, and the machinery degrades along the safe axis whenever the vocabulary is wrong.

14.3 Functional decomposition of the route representation

§14 opened with a claim we have so far only argued: route-relative description (‘ n steps past the last GOOD’) is provenance of a path, not identity of its endpoints. The last experiment measures it. A witness compresses its route to the water into a purely *relational* song — a sequence of couplets (σ, e) , where σ is the landmark constellation signature and e the exact beat vector from the previous couplet — carrying *no global coordinates whatsoever*. A band-limited traveler who has never seen the water receives it. Four registered arms (bit codec fixed before the runs; 12 paired worlds):

- **Beats only** (the pure ‘ n steps from the last GOOD’): usable only by *assuming* the anchor. With a shared anchor it is exact (11/11, the negative control); with anchor disagreement it locks every time and is right *never* (11/11 fail-open mislocks, displaced by exactly the anchor offset). Beats do not carry identity — they parasitise on the identity of the anchor.
- **Signatures only:** re-identifies co-visited places but has no edges to walk — *zero* transports of the never-seen target. Nodes alone cannot reach the unseen.
- **Signatures + beats:** couplets matched mutually-uniquely, the beat chain checked for metric consistency between matched couplets (a loop-closure test), the target dead-reckoned from the last matched landmark. Coverage 1.0, every transport exact, zero fail-open — at 366 bits per song.
- **Full snapshots** (the machinery of §14): the information upper bound — same correctness at 5,690 bits.

All four registered predictions passed. The song is not a lighter snapshot; it is a different *kind* of object: identity lives in the nodes (landmarks), transport lives in the edges (beats), and their composition delivers full-snapshot transport at 6.4% of the bit cost while inheriting the fail-safe property of §14.2 — everything that goes wrong goes wrong as a refusal. (This 6.4% is the *pure song codec*: nodes and edges only. A deployed collective-memory protocol adds provenance, timestamps, world versions, and a utility certificate; the full-protocol payload ratio, measured in the companion formation paper, is correspondingly larger.)

15 Related work

Shared memory in agent collectives. Generative Agents Park et al. (2023) and multi-agent LLM frameworks share memory streams or context but expose no lock interface and no per-source decomposition of a merged belief, so provenance-conditioned completion is not measurable there. Vector memories (e.g. MemGPT Packer et al. (2023)) merge by embedding arithmetic; the merge is not source-decomposable, which blocks both φ and the masked-replay counterfactual.

Emergent communication. CommNet Sukhbaatar et al. (2016) and TarMAC Das et al. (2019) learn continuous channels; the companion paper shows such channels collapse stage observability. Warp requires an explicit materialization event, which these architectures do not expose.

Stigmergy. Pheromone-style coordination is an *anonymous* warp: evidence transfers without provenance. Our results quantify what anonymity costs (shared-bus warp share is maximal while its warp completion is minimal) and what provenance buys (trust gates, reservations, rollback).

Optimistic concurrency. The rollback primitive is Time Warp Jefferson (1985) transplanted from parallel simulation to memory-driven navigation: locks are optimistic computations, occupancy evidence is a straggler message, anti- M^* is the anti-message. To our knowledge this transplant is new.

Map merging and place recognition. Multi-robot SLAM treats inter-map data association as an estimation problem over metric features Cadena et al. (2016); visual place recognition Lowry et al. (2016) matches appearance. Our contribution is not a competing matcher but the measurement layer on top: provenance-conditioned completion through recovered frames, the fail-closed/fail-open asymmetry as a registered prediction, and law invariance under frame removal.

Route sharing. Stigmergy transfers paths anonymously (pheromone gradients); learned communication (e.g. CommNet Sukhbaatar et al. (2016)) transfers policies opaquely. Neither exposes an edge-provenance decomposition, so neither can state — let alone predict — where a transferred route breaks. The rupture predictor is Time Warp Jefferson (1985) thinking applied to memory decay: deterministic replay of a gate over calibrated kinematics.

Songlines. The narrative framing corresponds to a functional distinction in both directions: the song is a sequence (route warp), its places are recognised by meaning (semantic identity), it fades verse by verse (rupture law), and it protects the walker while it lasts (hazard stratification).

16 Limitations

The provenance analysis is subject to six main limitations. First, the primary experiments use a shared coordinate frame and therefore do not solve cross-agent entity correspondence; the route-level analysis in the following sections addresses this assumption with semantic landmark matching. Second, the evidence is based mainly on controlled grid environments, with a directional VMAS study and a small text-world LLM demonstration rather than a photorealistic benchmark. Third, the sign of counterfactual warp gain is regime-dependent and should not be summarized as a universal benefit of foreign memory. Fourth, the closed-form distance law is specific to the explicit exponential trust–staleness gate and its pruning schedule. Fifth, the co-lock-pressure analysis is exploratory and is reported as mechanism evidence rather than as a confirmed predictive law. Sixth, the LLM experiment uses a limited model and task family. All deviations from the registered design and all failed predictions are retained in Appendix A.

The route-transfer experiments use controlled navigation substrates and a predefined semantic vocabulary. Place correspondence is tested under translations and quarter-turn rotations; this setting should be described as discrete frame recovery rather than arbitrary continuous $SE(2)$ estimation. The semantic fingerprints are sufficiently rich to support exact correspondence in the reported environments, but open-world appearance variation, perceptual uncertainty, and learned symbol alignment remain unresolved. The rupture law is exact for the specified edge-wise trust–staleness mechanism and does not by itself provide a universal safety guarantee. The claim concerning risk is therefore restricted to preservation of the witness’s measured hazard profile within the validated route prefix. Finally, the 6.4% figure refers to the pure route representation; a complete deployment cost must additionally include provenance, timestamps, correspondence hypotheses, and coordination metadata.

17 Conclusion

Taken together, the two movements of this paper give a single answer to what makes foreign navigation memory executable. The first showed that a peer-supported *commitment* carries value only under a provenance, validity, and coordination contract — direct completion understates it, a trust–staleness horizon bounds it, and a reservation protocol makes fast sharing net-positive. The second identifies a provenance-bearing relational route as the transferable unit of executable navigation memory. A target record by itself can identify a destination without specifying a feasible path, and an edge sequence without a reliable landmark correspondence can transport the agent toward the wrong referent. The experiments show that these functions are distinct: semantic landmarks provide identity, route edges provide transport, and only their composition supports transfer to an unseen target.

The route representation also exposes a temporal boundary. By applying the trust–staleness gate to each edge, the method predicts the route prefix that remains admissible and the edge at which the transfer should be rejected or replanned. In the controlled studies, this rupture index matches the observed failure location exactly. Meaning-based landmark correspondence further removes the shared coordinate assumption for discrete translations and quarter-turn rotations and fails closed when the observations are ambiguous.

The results support a precise interpretation of foreign navigation memory: it must preserve referential identity, transport structure, provenance, and a validity boundary together. How such memory is *formed*, consolidated, and admitted into a collective over long horizons — with explicit merge, trust, and staleness rules — is the subject of a companion paper on utility-certified collective memory.

A Deviations from the pre-registered design

1. Reservation tie-break extended to (lock tick, *distance at lock*, agent id): a pure id tie-break hands simultaneous locks to a distant agent over the target’s near-discoverer.
2. Protocol recovery is measured against the scarcity ceiling $\min(m, M)/m$ rather than the $K=8$ reference, separating structural from coordination collapse; unconditional metrics are reported alongside (Table 5).
3. Completion breakpoints use the discrete-time refinement of §5 (law constants unchanged; hold-out registered).
4. Collision pressure is operationalized as co-lock count rather than co-recipients (structurally constant $N-2$ under broadcast).
5. LLM demo: layered execution (LLM semantics, deterministic locomotion); completion attributed at the affordance radius.

B Reproducibility

All experiments run on CPU in minutes-to-tens-of-minutes; the LLM demo uses a local Ollama model with deterministic caching.

```
PYTHONPATH=. python experiments/warp/exp_warp_anchor.py
PYTHONPATH=. python experiments/warp/exp_warp_gain.py --seeds 20
PYTHONPATH=. python experiments/warp/analyze_warp_chains.py
PYTHONPATH=. python experiments/warp/exp_warp_gain_individual.py
PYTHONPATH=. python experiments/warp/exp_warp_age_law.py
PYTHONPATH=. python experiments/warp/exp_warp_age_law_holdout.py
PYTHONPATH=. python experiments/warp/exp_warp_drive.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_cross_session.py --layouts 10

PYTHONPATH=. python experiments/warp/exp_route_warp_r0.py
PYTHONPATH=. python experiments/warp/exp_route_warp_r1.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_route_warp_r2.py --seeds 5
PYTHONPATH=. python experiments/warp/exp_route_warp_r3.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_semantic_identity.py
PYTHONPATH=. python experiments/warp/exp_warp_semantic_stack.py --seeds 20
PYTHONPATH=. python experiments/warp/exp_warp_rotation.py --seeds 10
PYTHONPATH=. python experiments/warp/exp_warp_landmark_ablation.py
PYTHONPATH=. python experiments/song_grammar/exp_s0_song_smoke.py
```

C Full counterfactual sign map

D Registered predictions and revisions

All predictions were written to disk before execution; the artifact retains both versions wherever an operationalisation was revised. Revisions: R1 negative control re-measured in path length; R1 gain monotonicity re-formulated as cliff detection; W9 full-stack discriminator changed from success to warp-lock precision; W8 strict- W^* functioning extended to include warp-assisted own completions (taxi chains). The vocabulary ablation (A1–A4, §14.2) and the song-anatomy experiment (S0.1–S0.4, §14.3, bit codec registered with the predictions) passed as registered, with no revisions.

References

- C. Packer, V. Fang, S. Patil, K. Lin, S. Wooders, and J. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv:2310.08560*, 2023.
- J. S. Park, J. C. O’Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. *arXiv:2304.03442*, 2023.
- S. Sukhbaatar, A. Szlam, and R. Fergus. Learning multiagent communication with backpropagation. In *NeurIPS*, 2016.
- A. Das, T. Gervet, J. Romoff, D. Batra, D. Parikh, M. Rabbat, and J. Pineau. TarMAC: Targeted multi-agent communication. In *ICML*, 2019.
- D. R. Jefferson. Virtual time. *ACM Transactions on Programming Languages and Systems*, 7(3):404–425, 1985.
- C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard. Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age. *IEEE Transactions on Robotics*, 32(6):1309–1332, 2016.
- S. Lowry, N. Sünderhauf, P. Newman, J. J. Leonard, D. Cox, P. Corke, and M. J. Milford. Visual place recognition: A survey. *IEEE Transactions on Robotics*, 32(1):1–19, 2016.